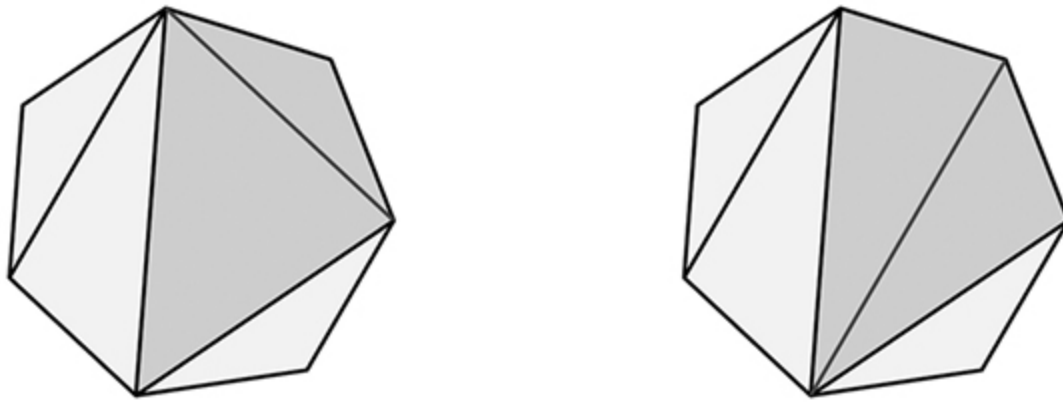


The `points_to_triangle()` function we defined near the beginning of the chapter is the simplest possible triangulation algorithm. However, it's quite limited because it works only if we give it exactly three input points. If we want to triangulate three points, there's only one possible way to do so: output a triangle consisting of exactly those three points. If we want to triangulate more than three points, there will inevitably be more than one way to triangulate. For example, consider the two distinct ways to triangulate the same seven points shown in [Figure 7-8](#).



[Figure 7-8](#): Two different ways to triangulate seven points (Wikimedia Commons)

In fact, there are 42 possible ways to triangulate this regular heptagon ([Figure 7-9](#)).